Recording L01-R01
Quadratic separation of a Gaussian

A proof of the formula for the conditional density of a Gaussian

Probabilistic Machine Learning 2027-1

2026-10-17

From the lecture to this proof

From the lecture L01: Gaussian blocks, conditioning and centered residual

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_d(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$\Sigma = \Sigma^T \succ 0, \quad d = d_1 + d_2$$

To prove here

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; m(x_2), S)$$

The conditional mean $m(x_2)$ and covariance S obtained in the lecture. Derive this complete density from the supplied joint density $p(x_1, x_2)$.

Fix the observed value x_2; the density describes the missing block X_1.
We account for the exponential and its coefficient, then identify the conditional by integration and division.

The density contains an exponential and a coefficient

$$p(x_1, x_2) = C \exp\left(-\frac{1}{2} Q\right)$$

The quadratic in the exponent

$$Q = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

Target: a residual contribution plus an observed-deviation contribution.

Residual: $x_1 - m(x_2)$. Observed deviation: $x_2 - \mu_2$.

The coefficient before the exponential

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

$$d = d_1 + d_2$$

Target: the coefficient for the residual density times that for the observed-block density.

> Writing the exponential as a product will not establish a full density product until the coefficient also matches.

Choose B to remove the cross covariance

Subtract B times the observed block so the cross covariance cancels:

$$H := \Sigma_{22}$$

$$\Sigma_{12} - BH = 0 \quad \Leftrightarrow \quad BH = \Sigma_{12}$$

Why the inverse of H exists

Take any nonzero vector z for the observed block:

$$z \in \mathbb{R}^{d_2} \setminus \{0\}$$

$$z^T H z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0$$

$$H = H^T \succ 0 \quad \Rightarrow \quad H^{-1} \text{ exists}$$

Solve the cancellation equation

$$B = \Sigma_{12} H^{-1} \in \mathbb{R}^{d_1 \times d_2}$$

$$BH = \Sigma_{12}, \quad HB^T = \Sigma_{21}$$

Canceling the cross blocks makes the inverse act separately on residual and observed coordinates. This is what will give the two terms in Q.

Subtracting the predicted deviation produces the residual

$$u := x_2 - \mu_2$$

$$m(x_2) := \mu_1 + Bu, \quad r := x_1 - m(x_2)$$

$$T := \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}$$

$$T(x - \mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

Target from the lecture — to prove here:

$$Q = r^T S^{-1} r + u^T H^{-1} u$$

> Here r is the error around the predicted mean and u is the observed deviation. The target uses the residual covariance S from the lecture; it will be derived here.

The coordinate change is reversible

The map sent $(x_1 - \mu_1, u)$ to (r, u); add the predicted deviation back:

$$U := \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \quad U \begin{pmatrix} r \\ u \end{pmatrix} = \begin{pmatrix} r + Bu \\ u \end{pmatrix} = x - \mu$$

$$TU = \begin{pmatrix} I & B - B \\ 0 & I \end{pmatrix} = I$$

$$UT = \begin{pmatrix} I & -B + B \\ 0 & I \end{pmatrix} = I \quad \Rightarrow \quad T^{-1} = U$$

> The residual and observation together determine the original centered vector. Invertibility will preserve strict positivity and let us recover the inverse covariance.

Subtract the observed rows to cancel the upper-right block

Using $BH = \Sigma_{12}$, multiply by T on the left:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

$$T\Sigma = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}$$

$$BH - BH = 0$$

$$S := \Sigma_{11} - BHB^T, \quad T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}$$

> The chosen B cancels the upper-right block exactly. The lower-left block remains because only the row operation has been applied.

The matching column operation cancels the lower-left block

Carry the preceding product forward and multiply by T^T on the right:

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}, \quad T^T = \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix} \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$= \begin{pmatrix} S & 0 \\ HB^T - HB^T & H \end{pmatrix}$$

$$D := T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

> The row and column operations together give the covariance in residual and observation coordinates. Its off-diagonal blocks are zero; its inverse will produce the two quadratic terms.

The remaining residual block is positive definite

The covariance congruence uses an invertible T and a supplied $\Sigma \succ 0$:

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

$$w \neq 0 \quad \Rightarrow \quad T^T w \neq 0$$

$$w^T D w = \left(T^T w\right)^T \Sigma \left(T^T w\right) > 0$$

$$a \in \mathbb{R}^{d_1} \setminus \{0\}, \quad w = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a^T S a = w^T D w > 0 \quad \Rightarrow \quad S = S^T \succ 0$$

The supplied joint strict positivity survives the reversible coordinate change. Both S and H therefore have inverses and positive determinants, as the Gaussian density requires.

Invert the covariance identity to reach the exponent

The density uses Σ^{-1}, while elimination established an identity for Σ:

$$D = T\Sigma T^T, \quad D = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

$$D^{-1} = T^{-T}\Sigma^{-1}T^{-1}$$

$$T^{-T} := \left(T^{-1}\right)^T$$

$$T^T D^{-1} T = \Sigma^{-1}$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} T$$

Invert the product in reverse order, then multiply on the left by T^T and on the right by T. Do not reorder the factors. This gives the inverse covariance needed in Q.

The joint quadratic becomes a residual term plus an observed term

From the covariance congruence and residual coordinates:

$$\Sigma^{-1} = T^T D^{-1} T, \quad T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}$$

Substitute into the quadratic Q from the joint density:

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu) = [T(x-\mu)]^T D^{-1} [T(x-\mu)]$$

$$= \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$

$$Q = \underbrace{r^T S^{-1} r}_{\text{residual term}} + \underbrace{u^T H^{-1} u}_{\text{observed term}}$$

At fixed x_2, u is fixed while $r = x_1 - m(x_2)$ varies with x_1. The zero off-diagonal blocks remove the mixed term.

A sum in the exponent becomes a product

Use the proved quadratic split in $\exp\left(-\frac{Q}{2}\right)$:

$$Q = r^T S^{-1} r + u^T H^{-1} u$$

Use $\exp(a+b) = \exp(a)\exp(b)$ with the quadratic split:

$$\exp\left(-\frac{1}{2}Q\right) = \exp\left(-\frac{1}{2}r^T S^{-1} r\right)\exp\left(-\frac{1}{2}u^T H^{-1} u\right)$$

$$p(x_1, x_2) = C\exp\left(-\frac{1}{2}r^T S^{-1} r\right)\exp\left(-\frac{1}{2}u^T H^{-1} u\right)$$

$$C = (2\pi)^{-\frac{d}{2}}(\det\Sigma)^{-\frac{1}{2}}$$

> The two exponentials have the required shapes. To make two complete Gaussian densities, C must supply the coefficient for each dimension and covariance.

The same elimination gives the determinant product

Reuse the same covariance congruence; T is block triangular with identity diagonal blocks:

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad \det T = \det I_{d_1} \det I_{d_2} = 1$$

$$\det D = \det T \det \Sigma \det T^T = \det \Sigma$$

$$\det D = \det S \det H \quad \Rightarrow \quad \det \Sigma = \det S \det H$$

The coefficient C contains $(\det \Sigma)^{-\frac{1}{2}}$. This identity replaces it by determinant factors for the residual and observed blocks.

The factors before the exponentials split too

The dimensions and covariance determinants both split; S and H are positive definite:

$$d = d_1 + d_2, \quad \det \Sigma = \det S \det H$$

$$(2\pi)^{-\frac{d}{2}} = (2\pi)^{-\frac{d_1}{2}} (2\pi)^{-\frac{d_2}{2}}$$

$$(\det \Sigma)^{-\frac{1}{2}} = (\det S)^{-\frac{1}{2}} (\det H)^{-\frac{1}{2}}$$

$$C = C_1 C_2$$

$$C_1 := (2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}}$$

$$C_2 := (2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}}$$

> The residual exponential receives C_1, for dimension d_1 and covariance S. The observed exponential receives C_2, for dimension d_2 and covariance H.

Reassemble the complete Gaussian densities

Gaussian density in x_1 for fixed x_2

$$g_1(x_1; x_2) := C_1 \exp\left(-\frac{1}{2} r^T S^{-1} r\right)$$

$$= \mathcal{N}_{d_1}(x_1; m(x_2), S)$$

Gaussian density in the observed block

$$g_2(x_2) := C_2 \exp\left(-\frac{1}{2} u^T H^{-1} u\right)$$

$$= \mathcal{N}_{d_2}(x_2; \mu_2, H), \quad u = x_2 - \mu_2$$

$$m(x_2) = \mu_1 + Bu, \quad r = x_1 - m(x_2)$$

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

Each coefficient and positive definite covariance matches its dimension. Integration and division will identify which factor is the marginal and which is the conditional.

Integrate over x1 to identify the marginal

Carry the product forward. For fixed x_2, $m(x_2)$ is fixed and $S \succ 0$:

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = 1$$

$$p(x_2) = \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1$$

$$= g_2(x_2) \int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = g_2(x_2)$$

$$p(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H)$$

For each fixed observation, the first factor integrates to one. Marginalizing the joint therefore leaves exactly the second factor.

Divide by the marginal to obtain the conditional density

The marginal is positive for every observed value:

$$p(x_2) = g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) > 0$$

$$p(x_1|x_2) = \frac{p(x_1, x_2)}{p(x_2)}$$

$$= \frac{g_1(x_1; x_2) g_2(x_2)}{g_2(x_2)} = g_1(x_1; x_2)$$

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

The factor identified by division is already normalized in x_1. This establishes the complete conditional density for every observed value x_2.

The observation can shift the conditional mean

From the proved conditional density:

$$m(x_2) = \mu_1 + B(x_2 - \mu_2), \quad \Sigma_{1|2} = S$$

$$m(x_2 + \Delta) - m(x_2) = B\Delta$$

$$S = \Sigma_{11} - \Sigma_{12} H^{-1} \Sigma_{21}$$

The mean can change with x_2 through B; S does not depend on the observed value.
The residual $r = x_1 - m(x_2)$ uses the observation. Its quadratic being separate does not make the original X_1 and X_2 independent.
The result is exact for the supplied joint Gaussian with $\Sigma \succ 0$; it does not test that model against data.